Customer Shopping Behavior Analysis: Project Report

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Executive Summary

This project analyzes 3,900 retail transactions worth $233,081 to show which customers, products and promotions drive revenue. It runs end to end: Python cleans the data, PostgreSQL answers 10 business questions, and Power BI presents the results in an interactive dashboard.

Headline findings

- **Revenue is volume-driven, not price-driven.** Men generate 67.7% of revenue ($157,890) because they are 68% of customers. Average spend per order is almost identical across genders ($59.54 vs $60.25).
- **Subscriptions are not yet paying off.** Subscribers (27% of customers) spend $59.49 per order against $59.87 for non-subscribers, so the program is not lifting basket size.
- **Discounts are heavily used.** 43% of orders carry a discount, and the top five products are discounted on 47% to 50% of their sales.
- **Clothing and Accessories carry the business.** Together they bring in 76.5% of revenue.
- **The base is already loyal.** 80% of customers have more than 10 previous purchases, so growth depends on order value and retention, not on acquiring new buyers.

Impact: the analysis turns raw transaction data into five actions covering subscription design, discount strategy, loyalty rewards, shipping and category focus (Section 7).

Business Problem and Objectives

A retail business holds detailed purchase records but lacks a clear view of who its best customers are, which products and promotions work, and where to focus marketing spend. This project answers those questions with data.

Objectives

1. Clean and prepare raw customer shopping data for reliable analysis.
2. Engineer features (age groups, purchase frequency, customer segments) that support segmentation.
3. Answer 10 business questions in SQL covering revenue, discounts, subscriptions, shipping and loyalty.

4. Build an interactive Power BI dashboard so non-technical stakeholders can explore the results.
5. Turn the findings into specific, actionable recommendations.

Dataset and Tech Stack

The dataset has 3,900 rows and 18 columns, one row per customer purchase, covering demographics, product details, payment and shipping choices, and loyalty indicators.

Area	Columns
Customer	Customer ID, Age, Gender, Location (50 US states), Subscription Status
Product	Item Purchased (25 items), Category (4), Size, Color, Season
Transaction	Purchase Amount (USD), Review Rating, Payment Method, Shipping Type
Behavior	Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases

Tool	Role in the project
Python (pandas)	Data cleaning, feature engineering, loading data into the database
SQLAlchemy and psycopg2	Connection between Python and PostgreSQL
PostgreSQL	Storage and business analysis with SQL
Power BI	Interactive dashboard and visual storytelling

Data Cleaning and Feature Engineering

The Python notebook prepares the data in six steps before it is loaded into PostgreSQL.

1. **Profile the data** with `info()`, `describe()` and null checks. Only Review Rating had gaps: 37 of 3,900 values (0.95%).
2. **Impute missing ratings** with the median rating of each product category, which keeps category-level differences intact.

3. **Standardize column names** to lowercase snake_case, for example `Purchase Amount (USD)` became `purchase_amount`, so SQL queries stay clean.
4. **Create `age_group`** using quartiles: Young Adult (18 to 31), Adult (32 to 44), Middle-aged (45 to 57) and Senior (58 to 70), giving four equal-sized groups.
5. **Create `purchase_frequency_days`** by converting labels such as Weekly, Monthly and Quarterly into numbers of days (7, 30, 90 and so on) for numeric analysis.
6. **Remove a redundant column.** `promo_code_used` matched `discount_applied` in every row, so it was dropped to avoid duplicate information.

The cleaned table of 19 columns was then written to a PostgreSQL table named `customer` through SQLAlchemy.

SQL Analysis: 10 Business Questions

Each question was solved in PostgreSQL using aggregations, subqueries, CASE logic, CTEs and window functions (`ROW_NUMBER() OVER (PARTITION BY ...)`).

#	Business question	Finding
1	Revenue by gender	Men $157,890 (67.7%), women $75,191 (32.3%). Men are 68% of customers; spend per order is nearly equal.
2	Discount users who spent above average	839 customers used a discount and still spent at or above the $59.76 average.
3	Top 5 products by average rating	Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78).
4	Standard vs Express shipping	Express orders average $60.48 against $58.46 for Standard (+3.4%).
5	Do subscribers spend more?	No. 1,053 subscribers average $59.49 and bring $62,645; 2,847 non-subscribers average $59.87 and bring $170,436.
6	Products with the highest discount rate	Hat 50.0%, Sneakers 49.7%, Coat 49.1%, Sweater 48.2%, Pants 47.4%.
7	New, Returning and Loyal segments	Loyal (over 10 purchases) 3,116 (79.9%), Returning (2 to 10) 701 (18.0%), New 83 (2.1%).

#	Business question	Finding
8	Top 3 products in each category	Accessories: Jewelry, Sunglasses, Belt. Clothing: Pants, Blouse, Shirt. Footwear: Sandals, Shoes, Sneakers. Outerwear: Jacket, Coat.
9	Are repeat buyers more likely to subscribe?	No. Of 3,476 customers with more than 5 purchases, only 958 (27.6%) subscribe, the same as the overall 27.0%.
10	Revenue by age group	Young Adult $62,143 (26.7%), Middle-aged $59,197, Adult $55,978, Senior $55,763.

Revenue by category (supporting the dashboard): Clothing $104,264 (44.7%), Accessories $74,200 (31.8%), Footwear $36,093 (15.5%), Outerwear $18,524 (7.9%).

Power BI Dashboard

The single-page Customer Behavior Dashboard lets a business user answer common questions in a few clicks, without writing SQL.

- **KPI cards:** Number of Customers, Average Purchase Amount and Average Review Rating, built with DAX measures.
- **Revenue views:** total revenue by product category and by age group.
- **Customer views:** customer count by category and by age group, plus a donut chart of the subscriber vs non-subscriber split.
- **Slicers:** Gender, Category, Subscription Status and Shipping Type filter every visual at once, so a manager can, for example, compare Express shipping among subscribed women in Accessories.

Together with the SQL results, the dashboard gives both a detailed and a visual view of the same customer data.

Key Insights and Recommendations

Five findings point to clear actions for the business.

Insight	Recommendation
Subscribers spend no more per order than non-subscribers ($59.49 vs $59.87).	Redesign the subscription with perks tied to spend, such as free shipping above a basket threshold or early access, and A/B test the result.
43% of orders use a discount, yet discounted orders average $59.28 against $60.13 without. Discounts are not raising basket size.	Reduce blanket discounts, especially on Hat, Sneakers and Coat (about 50% discounted), and use targeted offers instead.
72% of repeat buyers (2,518 of 3,476) are not subscribed.	Target these loyal customers with a subscription offer first, since they already trust the brand.
Express shipping orders are 3.4% larger than Standard.	Promote paid express upgrades at checkout.
Young Adults give the highest revenue (26.7%), and Clothing plus Accessories give 76.5%. Outerwear is only 7.9%.	Focus campaigns on these segments and categories, and review Outerwear range and seasonal promotion.

Limitations. The data is a single snapshot with no purchase dates, so trends over time cannot be measured. Differences in average spend between groups are small (1% to 3%) and were not tested for statistical significance, so they should be validated with an A/B test before large decisions.

Project Structure, Skills and Conclusion

```
customer-shopping-behavior-analysis/
├── data/customer_shopping_behavior.csv
├── notebooks/Customer_Shopping_Analysis.ipynb
├── sql/Customer.sql
├── dashboard/customer_behavior-dashboard.pbix
└── README.md
```

Skills demonstrated

- **Python:** pandas data cleaning, imputation, feature engineering, database loading with SQLAlchemy.
- **SQL (PostgreSQL):** analytical queries with CTEs, window functions, subqueries and conditional aggregation.
- **Power BI:** KPI design, DAX measures, slicers and interactive dashboards.

- **Business thinking:** turning numbers into recommendations a manager can act on.

Conclusion. The data shows a loyal customer base that spends a steady $60 per order whatever the gender, age, subscription or discount. Growth will therefore come from raising order value and converting loyal non-subscribers, not from deeper discounts. The pipeline is reusable: new data can be cleaned, queried and visualized with the same steps.